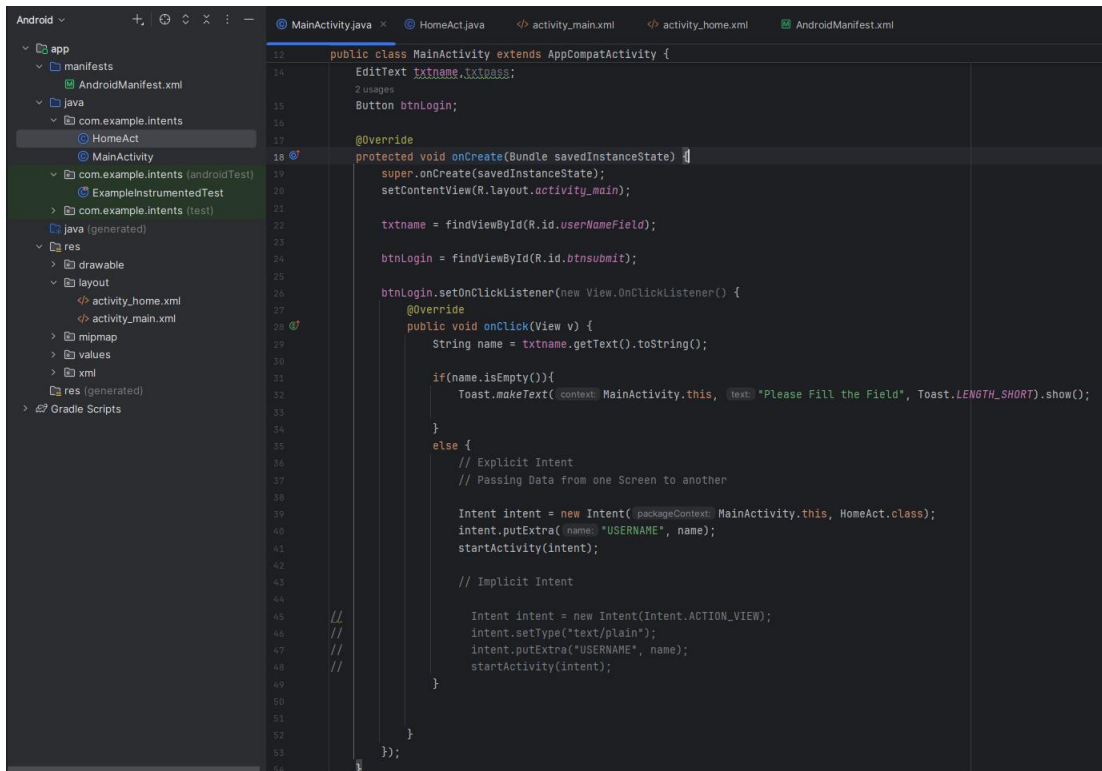


Mobile App Development Final Tyari (Codes)

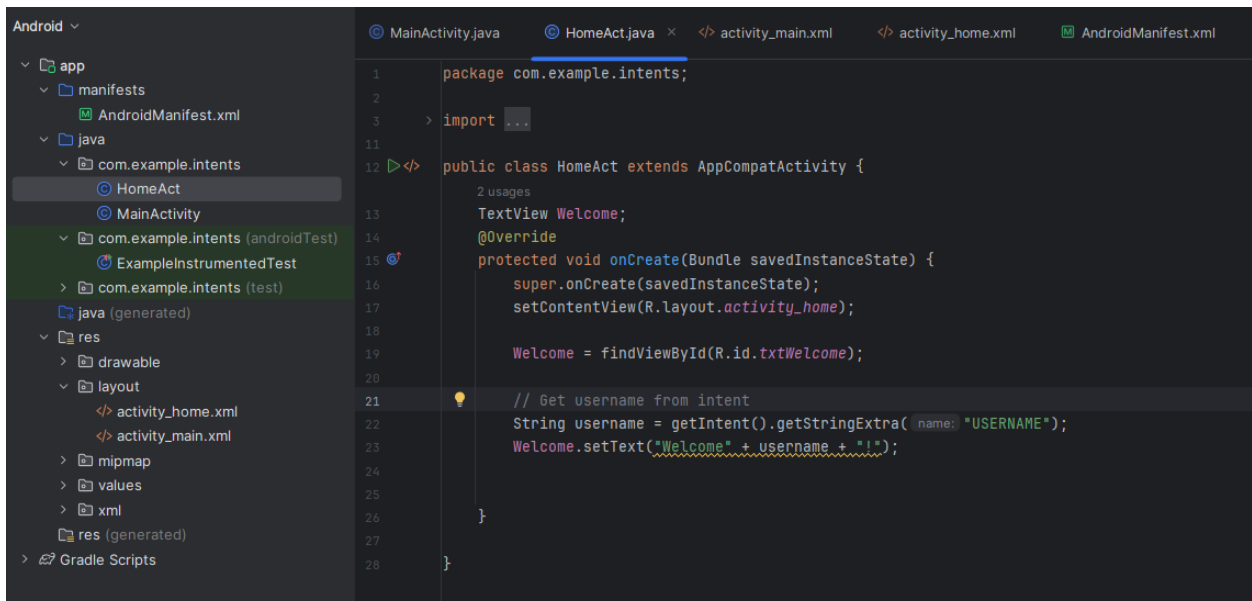
1- Intents, Passing data from one screen to another, explicit, implicit, Intent Filter

MainActivity.java



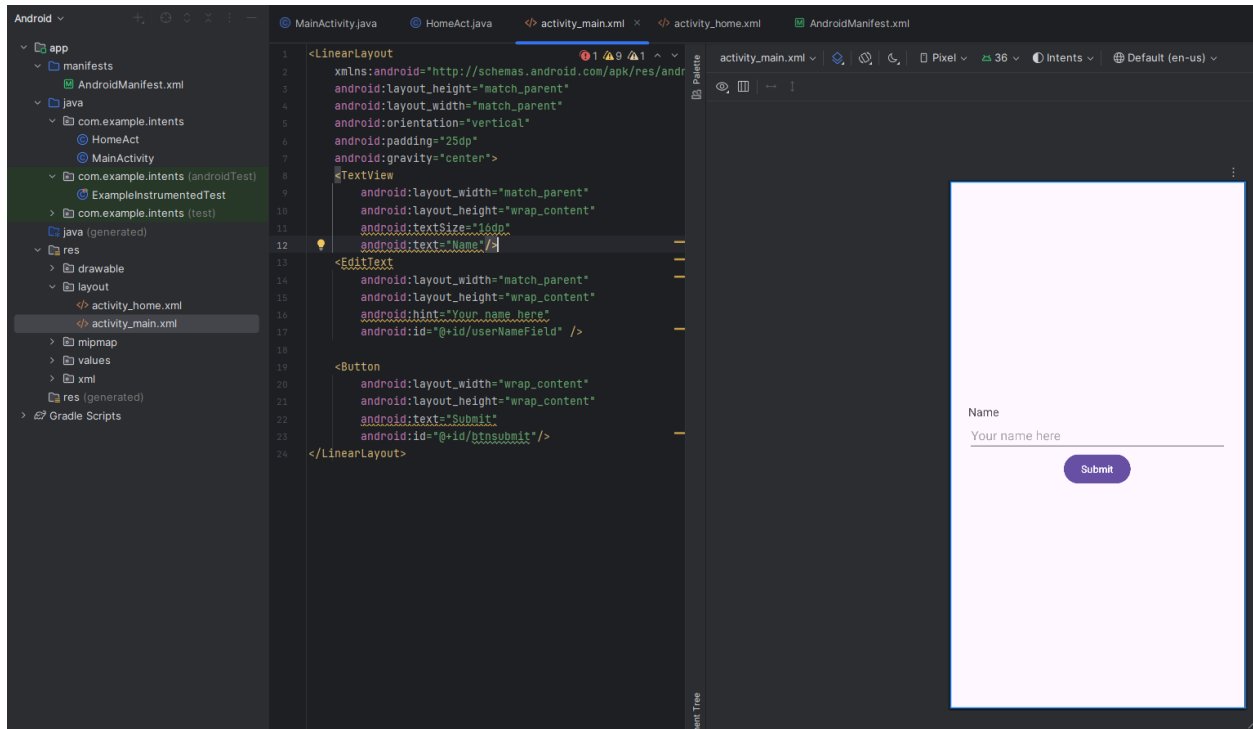
```
13 public class MainActivity extends AppCompatActivity {
14     EditText txtname,txtpass;
15     Button btnLogin;
16
17     @Override
18     protected void onCreate(Bundle savedInstanceState) {
19         super.onCreate(savedInstanceState);
20         setContentView(R.layout.activity_main);
21
22         txtname = findViewById(R.id.userNameField);
23
24         btnLogin = findViewById(R.id.btnsubmit);
25
26         btnLogin.setOnClickListener(new View.OnClickListener() {
27             @Override
28             public void onClick(View v) {
29                 String name = txtname.getText().toString();
30
31                 if(name.isEmpty()){
32                     Toast.makeText(context, MainActivity.this, "Please Fill the Field", Toast.LENGTH_SHORT).show();
33                 }
34                 else {
35                     // Explicit Intent
36                     // Passing Data from one Screen to another
37
38                     Intent intent = new Intent(MainActivity.this, HomeAct.class);
39                     intent.putExtra("name", name);
40                     startActivity(intent);
41
42                     // Implicit Intent
43
44                     Intent intent = new Intent(Intent.ACTION_VIEW);
45                     intent.setType("text/plain");
46                     intent.putExtra("USERNAME", name);
47                     startActivity(intent);
48                 }
49             }
50         });
51     }
52 }
53
54 }
```

HomeAct.java

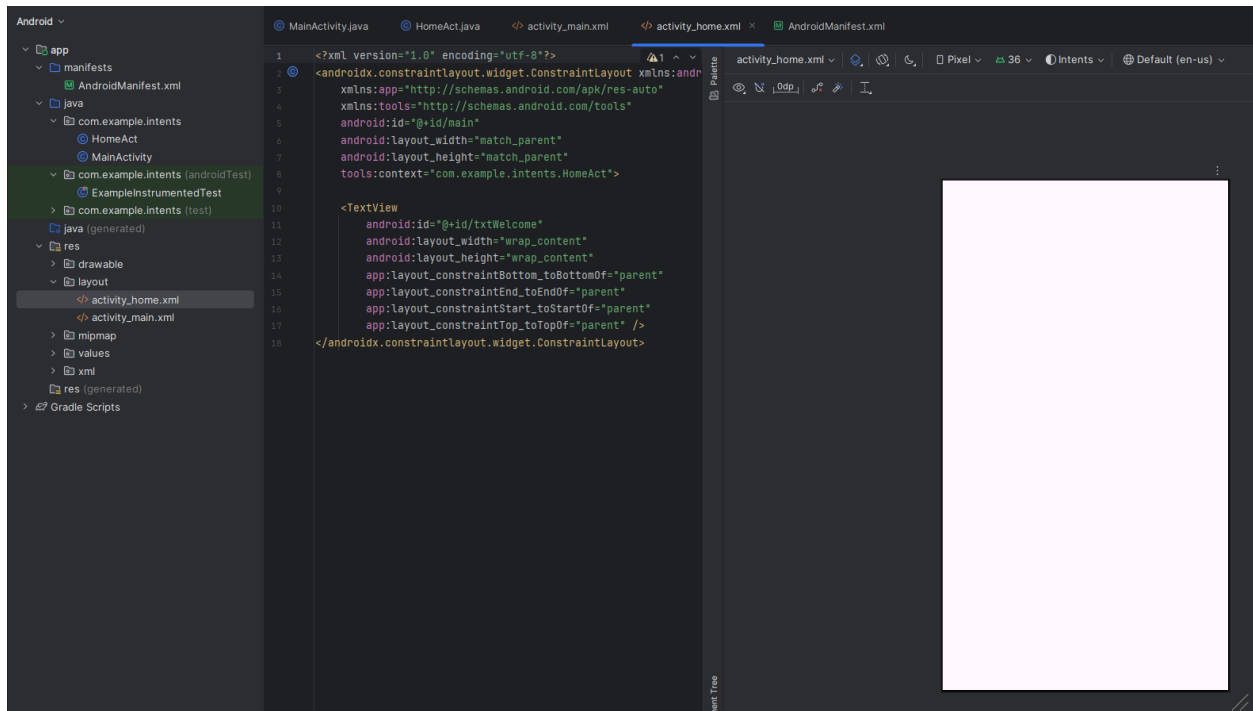


```
1 package com.example.intents;
2
3 import androidx.appcompat.app.AppCompatActivity;
4
5
6
7
8
9
10
11
12 public class HomeAct extends AppCompatActivity {
13     TextView Welcome;
14
15     @Override
16     protected void onCreate(Bundle savedInstanceState) {
17         super.onCreate(savedInstanceState);
18         setContentView(R.layout.activity_home);
19
20         Welcome = findViewById(R.id.txtWelcome);
21
22         // Get username from intent
23         String username = getIntent().getStringExtra("name");
24         Welcome.setText("Welcome" + username + "!");
25     }
26 }
27
28 }
```

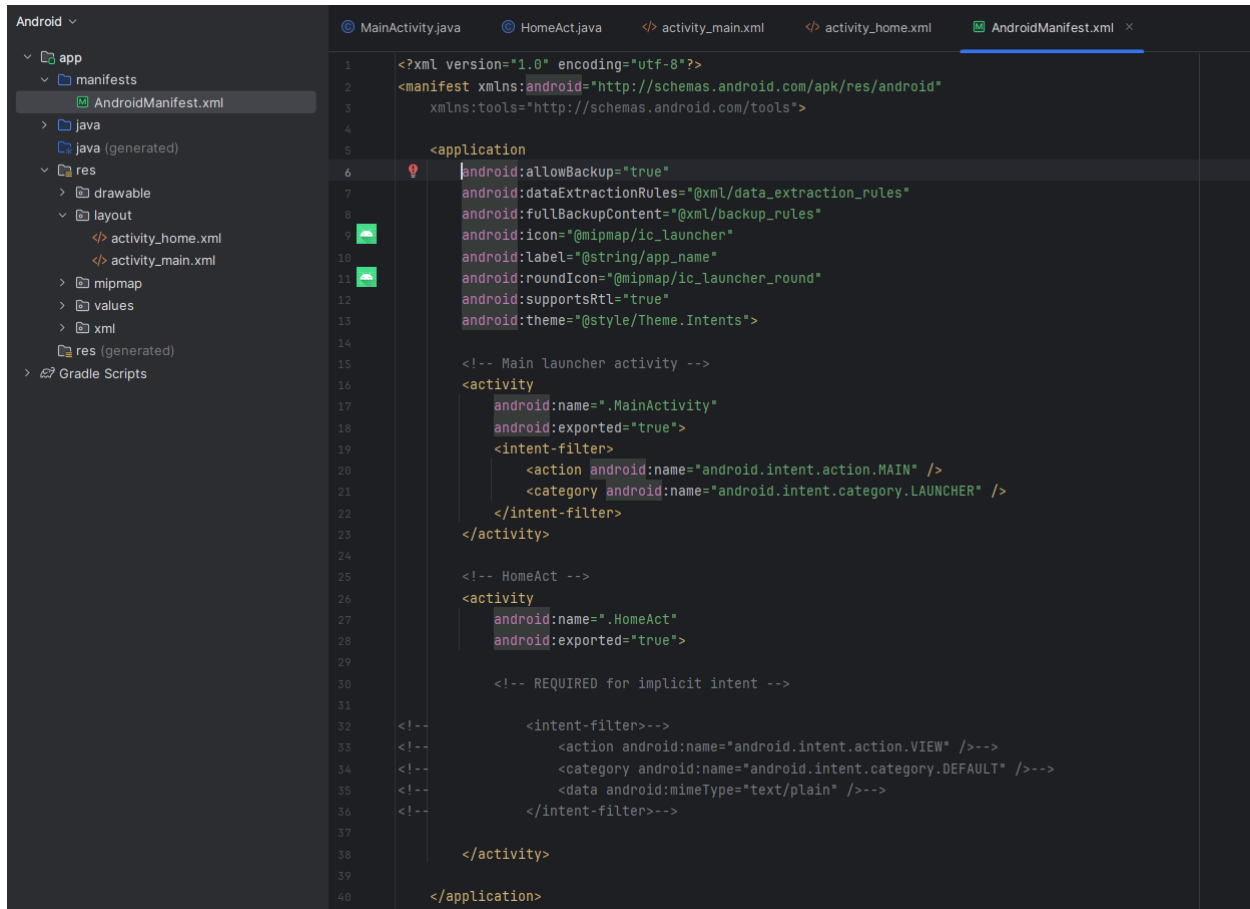
activity_main.xml



Activity_home.java



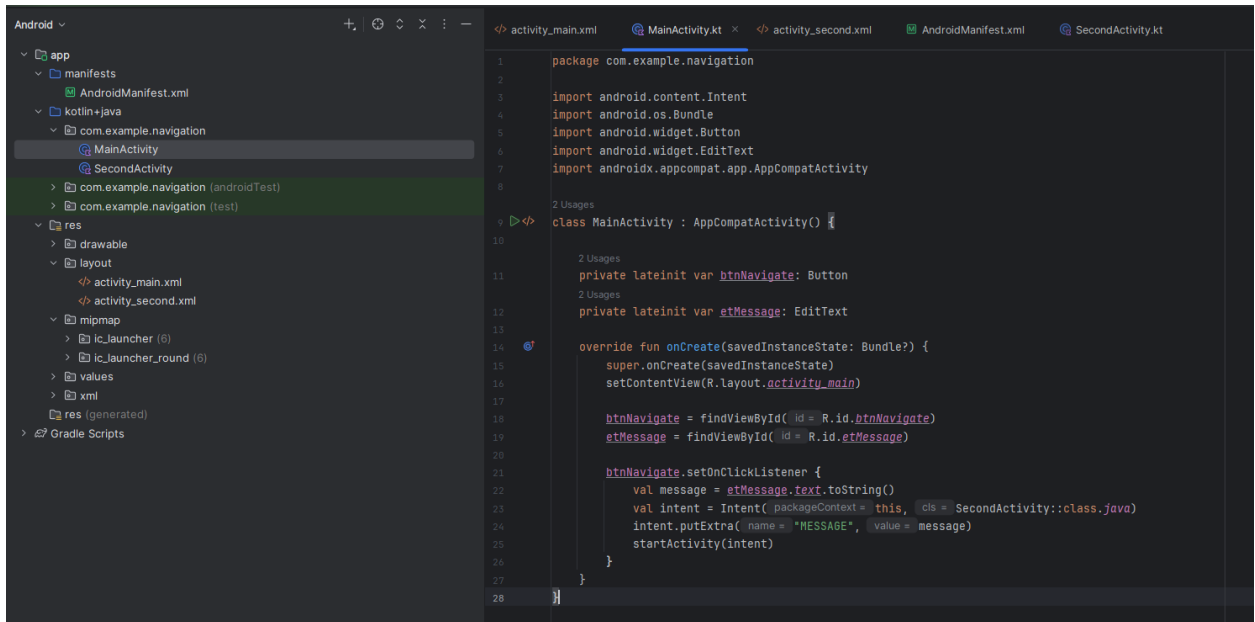
androidmanifest.xml



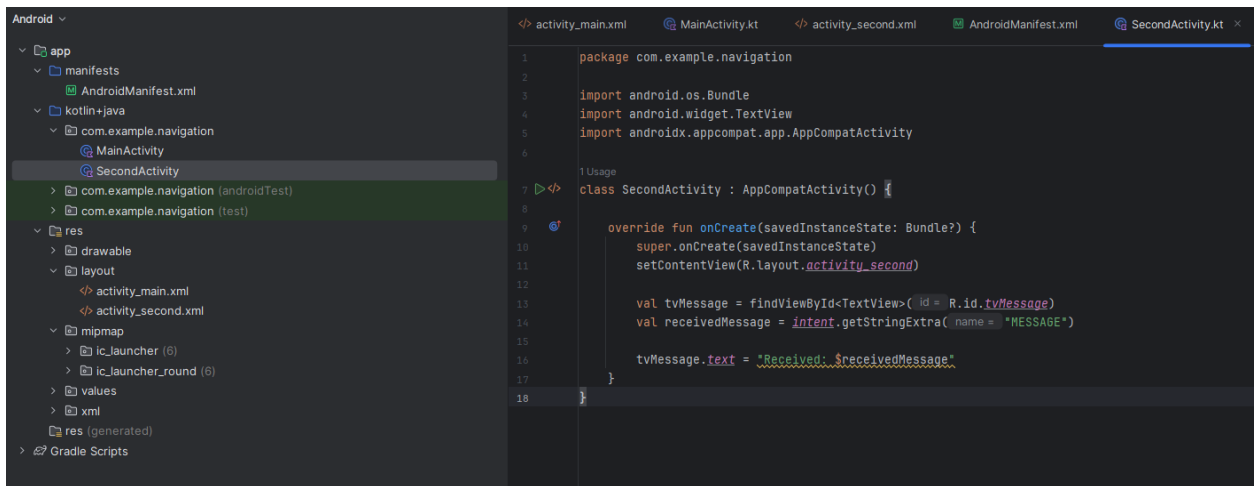
```
1 <?xml version="1.0" encoding="utf-8"?>
2 <manifest xmlns:android="http://schemas.android.com/apk/res/android"
3     xmlns:tools="http://schemas.android.com/tools">
4
5     <application
6         android:allowBackup="true"
7         android:dataExtractionRules="@xml/data_extraction_rules"
8         android:fullBackupContent="@xml/backup_rules"
9         android:icon="@mipmap/ic_launcher"
10        android:label="@string/app_name"
11        android:roundIcon="@mipmap/ic_launcher_round"
12        android:supportRtl="true"
13        android:theme="@style/Theme.Intents">
14
15        <!-- Main launcher activity -->
16        <activity
17            android:name=".MainActivity"
18            android:exported="true">
19            <intent-filter>
20                <action android:name="android.intent.action.MAIN" />
21                <category android:name="android.intent.category.LAUNCHER" />
22            </intent-filter>
23        </activity>
24
25        <!-- HomeAct -->
26        <activity
27            android:name=".HomeAct"
28            android:exported="true">
29
30            <!-- REQUIRED for implicit intent -->
31
32            <intent-filter>-->
33            <!--            <action android:name="android.intent.action.VIEW" />-->
34            <!--            <category android:name="android.intent.category.DEFAULT" />-->
35            <!--            <data android:mimeType="text/plain" />-->
36            <!--            </intent-filter>-->
37
38        </activity>
39
40    </application>
```

2- Navigation code (kotlin)

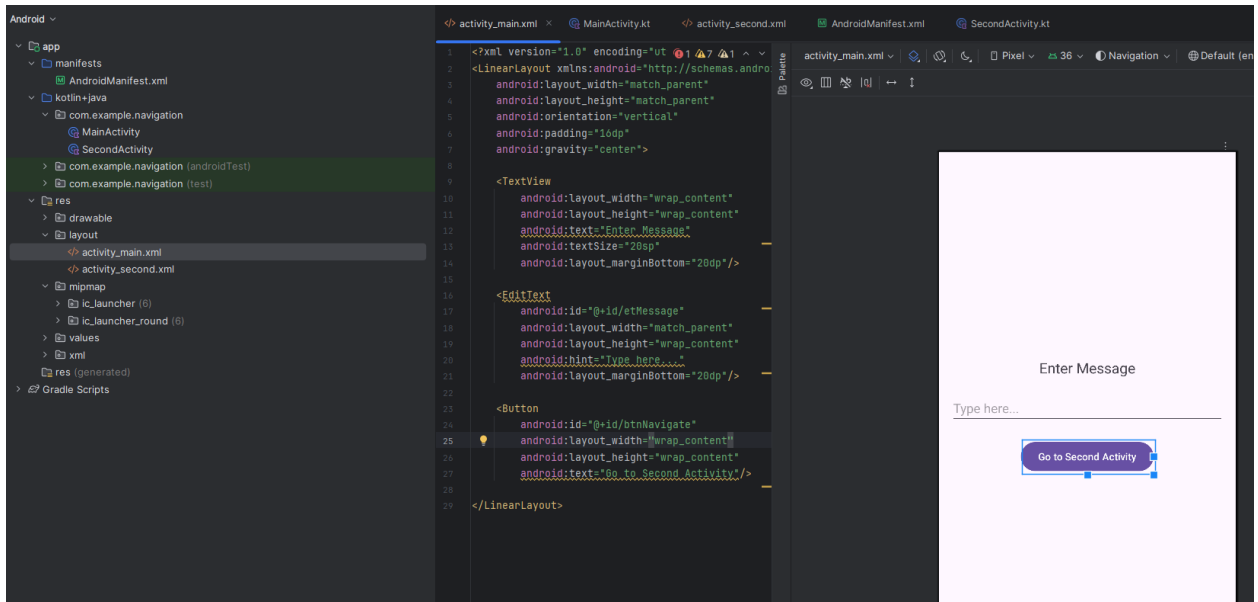
MainActivity.kt



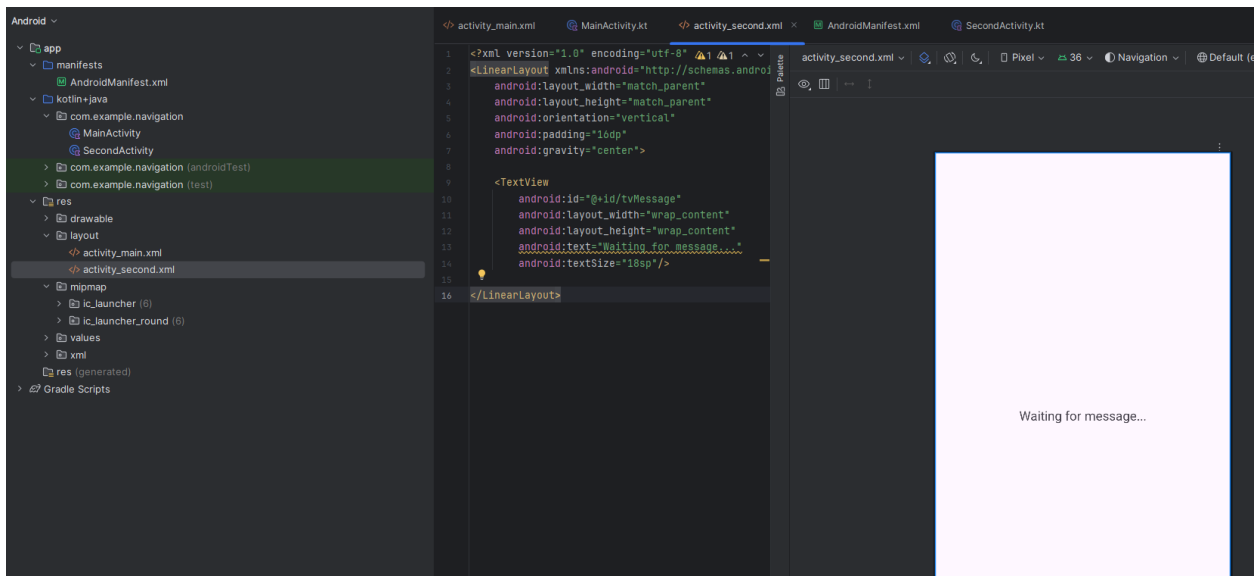
SecondActivity.kt



Activity_main.xml



Activity_second.xml



AndroidManifest.xml

```
2 <manifest xmlns:android="http://schemas.android.com/apk/res/android"
3         xmlns:tools="http://schemas.android.com/tools"
4         android:allowBackup="true"
5         android:fullBackupContent="true"
6         android:icon="@mipmap/ic_launcher"
7         android:label="@string/app_name"
8         android:roundIcon="@mipmap/ic_launcher_round"
9         android:supportRtl="true"
10        android:theme="@style/Theme.Navigation"
11        tools:targetApi="31">
12
13    <!-- MainActivity (Launcher Activity) -->
14    <activity
15        android:name=".MainActivity"
16        android:exported="true"
17        android:theme="@style/Theme.Navigation">
18        <intent-filter>
19            <action android:name="android.intent.action.MAIN" />
20            <category android:name="android.intent.category.LAUNCHER" />
21        </intent-filter>
22    </activity>
23
24    <!-- SecondActivity (Navigation Target) -->
25    <activity
26        android:name=".SecondActivity"
27        android:exported="false"
28        android:parentActivityName=".MainActivity">
29
30    </activity>
31
32    <!-- Add more activities here as needed -->
33    <!-- <activity android:name=".ThirdActivity" /> -->
34
35 </manifest>
```

3- Shared Preferences

Shared Preferences saves small data (like username, settings) so your app remembers it next time you open it

MainActivity.kt

```
activity_main.xml build.gradle.kts (SharedPreferences) build.gradle.kts (.app) MainActivity.kt x
11 class MainActivity : AppCompatActivity() {
12     // 9 Usages
13     companion object {
14         // 3 Usages
15         // SharedPreferences keys
16         const val PREF_NAME = "MyAppPrefs"
17         const val KEY_NAME = "name"
18         const val KEY_AGE = "age"
19         const val KEY_LOGGED_IN = "is_logged_in"
20     }
21
22     // 3 Usages
23     // Declare Views
24     private lateinit var etName: EditText
25     private lateinit var etAge: EditText
26     private lateinit var btnSave: Button
27     private lateinit var btnLoad: Button
28     private lateinit var btnClear: Button
29     private lateinit var tvData: TextView
30
31     override fun onCreate(savedInstanceState: Bundle?) {
32         super.onCreate(savedInstanceState)
33
34         // SET THIS LINE CORRECTLY
35         setContentView(R.layout.activity_main)
36
37         // 3 Usages
38         // Initialize views - Make sure IDs match XML exactly
39         etName = findViewById<EditText>(R.id.etName)
40         etAge = findViewById<EditText>(R.id.etAge)
41         btnSave = findViewById<Button>(R.id.btnSave)
42         btnLoad = findViewById<Button>(R.id.btnLoad)
43         btnClear = findViewById<Button>(R.id.btnClear)
44         tvData = findViewById<TextView>(R.id.tvData)
45
46         // Save button click
47         btnSave.setOnClickListener {
48             saveData()
49         }
50
51         // Load button click
52         btnLoad.setOnClickListener {
53             loadData()
54         }
55
56         // Clear button click
57         btnClear.setOnClickListener {
58             clearData()
59         }
60
61         // Load saved data when activity starts
62         loadData()
63     }
64
65     // 1 Usage
66     private fun saveData() {
67         val name = etName.text.toString().trim()
68         val ageText = etAge.text.toString().trim()
69
70         if (name.isEmpty() || ageText.isEmpty()) {
71             Toast.makeText(context = this, text = "Please fill all fields", duration = Toast.LENGTH_SHORT).show()
72             return
73         }
74
75         val age = ageText.toIntOrNull() ?: 0
76
77         val sharedPreferences = getSharedPreferences(PREF_NAME, Context.MODE_PRIVATE)
78         val editor = sharedPreferences.edit()
79
80         editor.putString(KEY_NAME, name)
81         editor.putInt(KEY_AGE, age)
82         editor.putBoolean(KEY_LOGGED_IN, true)
83         editor.apply()
84
85         Toast.makeText(context = this, text = "Data saved successfully!", duration = Toast.LENGTH_SHORT).show()
86     }
87 }
```

```

80
81 Toast.makeText(context = this, text = "Data saved successfully!", duration = Toast.LENGTH_SHORT).show()
82 clearFields()
83 }
84
85 2 Usages
86 private fun loadData() {
87     val sharedPreferences = getSharedPreferences(name = PREF_NAME, mode = Context.MODE_PRIVATE)
88
89     val name = sharedPreferences.getString(p0 = KEY_NAME, p1 = "Not set")
90     val age = sharedPreferences.getInt(p0 = KEY_AGE, p1 = 0)
91     val isLoggedIn = sharedPreferences.getBoolean(p0 = KEY_LOGGED_IN, p1 = false)
92
93     val result = """
94         Name: $name
95         Age: $age
96         Logged In: $isLoggedIn
97         Last Updated: ${java.text.SimpleDateFormat(pattern = "HH:mm:ss").format(date = java.util.Date())}
98     """.trimIndent()
99
100     tvData.text = result
101 }
102
103 1 Usage
104 private fun clearData() {
105     val sharedPreferences = getSharedPreferences(name = PREF_NAME, mode = Context.MODE_PRIVATE)
106     val editor = sharedPreferences.edit()
107     editor.clear()
108     editor.apply()
109
110     clearFields()
111     tvData.text = "All data cleared!"
112     Toast.makeText(context = this, text = "Data cleared successfully!", duration = Toast.LENGTH_SHORT).show()
113 }
114
115 2 Usages
116 private fun clearFields() {
117     etName.text.clear()
118     etAge.text.clear()
119 }

```

Activity_main.xml

The screenshot displays the Android Studio environment. On the left, the 'app' module is selected. The main editor shows the `activity_main.xml` file with the following XML content:

```

1 <?xml version="1.0" encoding="utf-8"?>
2 <LinearLayout xmlns:android="http://schemas.android.com/apk/res-auto"
3     xmlns:tools="http://schemas.android.com/tools"
4     android:layout_width="match_parent"
5     android:layout_height="match_parent"
6     android:orientation="vertical"
7     android:padding="16dp"
8     android:gravity="center"
9     tools:context=".MainActivity">
10
11     <TextView
12         android:layout_width="wrap_content"
13         android:layout_height="wrap_content"
14         android:text="SharedPreferences Demo"
15         android:textSize="24sp"
16         android:textStyle="bold"
17         android:layout_marginBottom="20dp"/>
18
19
20     <EditText
21         android:id="@+id/etName"
22         android:layout_width="match_parent"
23         android:layout_height="wrap_content"
24         android:hint="Enter your name"
25         android:layout_marginBottom="10dp"/>
26
27     <EditText
28         android:id="@+id/etAge"
29         android:layout_width="match_parent"
30         android:layout_height="wrap_content"
31         android:hint="Enter your age"
32         android:inputType="number"
33         android:layout_marginBottom="20dp"/>
34
35     <Button
36         android:id="@+id/btnSave"
37         android:layout_width="match_parent"
38         android:layout_height="wrap_content"
39         android:text="Save Data"
40         android:layout_marginBottom="10dp"/>

```

On the right, a preview of the app's UI is shown. It features a white background with a purple header bar. The title "SharedPreferences Demo" is centered at the top. Below the title, there are two input fields: "Enter your name" and "Enter your age". Underneath these fields are three purple buttons: "Save Data", "Load Data", and "Clear All Data". At the bottom, there is a light green box with the text "Your data will appear here...".

